Thresholding on ResNet18 Scores with Synthetic Negatives

Key Concepts: KO Role, Logit Computation, and "Max OK + 15%" Threshold

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Abstract

This document explains, in an operational and mathematically clean way: (i) the role of *fake KOs* (synthetic negatives) during training, (ii) how ResNet18 produces a numerical value (logit) and the associated "KO probability" for each image, (iii) how the decision threshold is computed in production using **only real OK images** with the rule:

$$\textbf{threshold} = \max(\text{score on OKs}) \cdot (1 + 0.15).$$

The core idea is to clearly separate two different problems: **learning the score** vs **calibrating the threshold**.

Contents

1 Setup and Notation

Let $x \in \mathcal{X}$ be an input image (after preprocessing: resize, normalization, masking, etc.). The model outputs a real-valued score through ResNet18.

1.1 Labels and Dataset

Let $y \in \{0, 1\}$ be the label:

$$y = 0 \iff x \text{ is OK}, \qquad y = 1 \iff x \text{ is KO}.$$

In real industrial scenarios we often have:

- many $\mathrm{OK}_{\mathrm{real}}$ images (real OKs),
- very few or no $\mathrm{KO}_{\mathrm{real}}$ images (real KOs).

To enable binary supervised training, we generate synthetic negatives:

$$x_{\mathrm{OK}} \xrightarrow[\text{inpainting / noise / blending / brightness}]{\text{transformations}} x_{\mathrm{KO}_{\mathrm{fake}}}.$$

2 How ResNet18 Assigns a Value to Each Image: The Logit

2.1 Definition: Logit

A standard ResNet18 (adapted for binary classification) computes:

$$h(x) \in \mathbb{R}^{512} \quad \text{(feature vector after global average pooling)}$$

and then a final fully connected layer produces:

$$z(x) = w^\top h(x) + b \in \mathbb{R},$$

where $w \in \mathbb{R}^{512}$ and $b \in \mathbb{R}$ are learned parameters.

This $z(x)$ is called the logit. It is not a probability; it is an unconstrained real number.

2.2 From Logit to "KO Probability" via Sigmoid

To map the logit into $(0, 1)$ we apply the sigmoid:

$$p_{\mathrm{KO}}(x) = \sigma(z(x)) = \frac{1}{1 + e^{-z(x)}}.$$

Example: if $z(x) = -7.2$, then

$$p_{\mathrm{KO}}(x) \approx \frac{1}{1 + e^{7.2}} \approx 7.4 \times 10^{-4}.$$

2.3 Important Fact: Order Invariance

The sigmoid function is strictly increasing:

$$z(x_1) < z(x_2) \iff \sigma(z(x_1)) < \sigma(z(x_2)).$$

Therefore, ranking images by logits or by probabilities is equivalent. Thresholds can be defined in either space.

3 Role of Fake KOs (Synthetic Negatives) During Training

3.1 Problem: Lack of Real KOs

Binary supervised learning requires both classes. If only OK images are available, the loss cannot learn a meaningful boundary.

3.2 Solution: Create a Synthetic Negative Class

We construct training pairs:

$$(x_{\mathrm{OK}_{\mathrm{real}}}, y = 0), \qquad (x_{\mathrm{KO}_{\mathrm{fake}}}, y = 1).$$

Each batch contains both classes, enabling gradient-based separation.

3.3 What the Model Actually Learns

The model is not learning the true distribution of real KOs. Instead, it learns to separate:

$$P(x \mid \mathrm{OK}_{\mathrm{real}}) \quad \text{vs} \quad P(x \mid \mathrm{KO}_{\mathrm{fake}}).$$

In practice, the network becomes a **guided anomaly scorer**:

- real OKs are pushed toward very negative logits ($z \ll 0$),
- fake KOs are pushed toward positive logits ($z \gg 0$).

If synthetic transformations are realistic, the learned features generalize to real defects even if distributions differ.

3.4 Why the Logit Depends on Fake KOs

The parameters w, b and the feature extractor $h(\cdot)$ are optimized using $(\mathrm{OK}_{\mathrm{real}})$ vs $(\mathrm{KO}_{\mathrm{fake}})$. Thus, fake KOs shape the score scale itself. They define the margin learned by the network.

4 Binary Cross Entropy and Logit Interpretation

4.1 Binary Cross Entropy (BCE)

With label $y \in \{0, 1\}$ and probability $p_{\mathrm{KO}}(x)$:

$$\mathcal{L}(x, y) = -y \log p_{\mathrm{KO}}(x) - (1 - y) \log(1 - p_{\mathrm{KO}}(x)).$$

Equivalently, BCE pushes:

- if $y = 0$ (OK): $p_{\mathrm{KO}}(x) \to 0$ and $z(x) \to -\infty$,
- if $y = 1$ (fake KO): $p_{\mathrm{KO}}(x) \to 1$ and $z(x) \to +\infty$.

4.2 Practical Interpretation

After training:

- typical OK: very negative logit $\Rightarrow$ tiny $p_{\mathrm{KO}}(x)$,
- fake KO: higher or positive logit $\Rightarrow$ large $p_{\mathrm{KO}}(x)$.

Even if $p_{\mathrm{KO}}(x)$ is not perfectly calibrated, it provides a reliable ranking score.

5 Threshold Computation: Option 3 (Max OK + 15%)

5.1 Principle: Calibrate on Real OKs Only

Threshold calibration must reflect real production data. Therefore, we use only $\mathrm{OK}_{\mathrm{real}}$ images.

5.2 Practical Pipeline

Given a validation/calibration set of only real OKs:

$$\{x_i\}_{i=1}^{n_0}, \quad y_i = 0 \ \forall i,$$

compute:

$$s_i = p_{\mathrm{KO}}(x_i) = \sigma(z(x_i)).$$

Then:

$$s_{\max} = \max_{i=1,\dots,n_0} s_i.$$

The Option 3 threshold is:

$$\boxed{\tau = s_{\max}\,(1+m), \quad m = 0.15}$$

i.e.

$$\boxed{\tau = 1.15 \cdot \max(\text{score on real OKs})}.$$

5.3 Decision Rule

For a new image x:

$$\hat{y}(x) = \begin{cases} 1 & \text{if } p_{\mathrm{KO}}(x) \geq \tau \quad \text{(predict KO)} \\ 0 & \text{if } p_{\mathrm{KO}}(x) < \tau \quad \text{(predict OK)} \end{cases}$$

5.4 Why "Max + Margin" Works

This rule is conservative:

- take the **worst OK** observed,
- add a safety margin (15%) to absorb noise, lighting changes, and minor domain shifts,
- anything exceeding this buffered maximum is considered anomalous.

5.5 Working on Logits vs Probabilities

The method above is multiplicative on probabilities. An additive margin on logits is also possible, but it changes the geometry. Consistency suggests working directly on p_{KO} if the rule is multiplicative.

6 Training vs Threshold: Clear Separation of Roles

Phase	**Inputs**	**Output**
Training	real OKs + fake KOs	scoring function $x \mapsto z(x)$ (and $p_{\mathrm{KO}}(x)$)
Threshold Calibration	ONLY real OKs	threshold $\tau = 1.15 \cdot \max s_i$
Production	new images	KO/OK decision using $p_{\mathrm{KO}}(x) \geq \tau$

Key Message. Fake KOs build the ruler (the score scale). The threshold places the red line on the ruler using real OK data.

7 Complete Numerical Example

Assume $n_0 = 32$ real OK validation images with scores:

$$\{s_i\} = [0.00010, 0.00012, 0.00009, \ldots, 0.00031].$$

Then:

$$s_{\max} = 0.00031, \qquad \tau = 1.15 \cdot 0.00031 = 0.0003565.$$

In production:

- if $p_{\mathrm{KO}}(x) = 0.00020 < \tau \Rightarrow$ OK,
- if $p_{\mathrm{KO}}(x) = 0.00150 \geq \tau \Rightarrow$ KO.

8 Operational Checklist

1. **Inference score:** compute $z(x)$ and $p_{\mathrm{KO}}(x) = \sigma(z(x))$.
2. **Training:** use $(\mathrm{OK}_{\mathrm{real}}, 0)$ and $(\mathrm{KO}_{\mathrm{fake}}, 1)$ to learn separation.
3. **Threshold:** use **only** $\mathrm{OK}_{\mathrm{real}}$ and set $\tau = 1.15 \cdot \max p_{\mathrm{KO}}(x)$.
4. **Decision:** KO if $p_{\mathrm{KO}}(x) \geq \tau$.
5. **Maintenance:** if lighting/camera/setup changes, recalibrate τ on new OK data.

9 Final Notes

- If strict FPR control is required, percentile-based thresholds provide theoretical guarantees. The "max + margin" rule is an empirical conservative heuristic.
- If calibrated probabilities are needed, apply post-hoc calibration (e.g., Platt scaling or isotonic regression) using labeled validation data.